PENGZHEN LI

University of Tennessee, Knoxville, Department of Economics, Cherokee Mills F154
+1 (865) 266–5815 | pli26@vols.utk.edu | Homepage: pengzhenli.github.io/industry/
LinkedIn | Google Scholar

SUMMARY

Final-year Economics Ph.D. candidate (expected May 2027) specializing in applied econometrics, causal inference, and optimization. Four peer-reviewed publications (one with 1,000+ citations), five working papers including a job market paper on trade restrictiveness with James Lake, 2024 Holly Award for research and the 2026 Garrison Award for teaching. Build reproducible analytics in Stata/R/Python/SQL/GAMS and turn large administrative, trade, and production datasets into decisions. Authorized to work in the U.S. without sponsorship.

EDUCATION

University of Tennessee, Knoxville Jan 2020 – May 2027 (expected)
Ph.D. in Economics — fields: international trade, applied econometrics, environmental and resource economics

Shandong University Sep 2016 – Jun 2018
M.A. in Finance — thesis published in *Technological Forecasting and Social Change* (1,000+ citations)

Tongji University, Shanghai Sep 2011 – Jun 2015
B.A. in Economics

SKILLS

Programming / software: Python (pandas, numpy, statsmodels, scikit-learn), R, Stata, SQL, GAMS, MATLAB, ArcGIS, IMPLAN, Git, LaTeX.
Methods: causal inference (DiD, event study, IV, RDD, synthetic control, fixed effects), panel and time-series econometrics (GMM, panel threshold, TVAR, VECM, mGARCH), forecasting, mixed-integer and stochastic optimization, DEA, game theory (Shapley value, mean-field games), supervised and unsupervised machine learning, text-as-data.
Credentials / languages: CFA Level II; English (fluent), Mandarin (native).

RESEARCH EXPERIENCE

Doctoral Researcher, Department of Economics, University of Tennessee 2023 – present

- *Trade restrictiveness under discriminatory tariffs* (with J. Lake; job market paper): derived welfare-based indices for partner-specific tariffs and a level/discrimination decomposition; built the WITS/Comtrade pipeline for 2017–2024 and quantified how MFN-only indices understate trade-war distortions. Presented at the Midwest Trade Meetings, 2025.
- *Multilateral tariff bargaining* (with G. Schaur): cooperative-game (Shapley value) model of coalition stability and gains from liberalization, extending Bagwell-Staiger-Yurukoglu (2021) to the multilateral case.
- *Mayoral endorsements and presidential elections* (with M. Van Essen, L. Lima): mean-field-game model of strategic endorsement timing tested on Brazilian and French administrative data; 2024 Holly Award.

Graduate Research Assistant, University of Tennessee Jan 2020 – Aug 2023
USDA NIFA and FAA ASCENT funded projects on sustainable-aviation-fuel supply chains.

- Built mixed-integer optimization models (GAMS) to design cost- and profit-optimal forest-residue sustainable-aviation-fuel supply chains across the Southeast U.S.; published in *Biomass and Bioenergy* (2026) and *Transportation Research Record* (2024).

- Ran facility-siting and regional impact analysis with IMPLAN and efficiency analysis with DEA; delivered scenario and sensitivity results (EMP, supervised ML, @RISK) to an interdisciplinary engineering–economics team and federal sponsors.
- Engineered reproducible ArcGIS/R/Python pipelines for high-resolution spatial data.

Earlier applied projects 2016 – 2020

- Panel-threshold evidence on green innovation and CO_2 (*TFSC* 2019, 1,000+ citations); synthetic-control estimate of the Wenchuan earthquake's long-run output effect; GMM analysis of MFP/CFAP payments and U.S. agricultural lending; TVAR/VECM/mGARCH commodity-price models; equity-premium prediction from *NYT*/*WSJ* text.

SELECTED PUBLICATIONS

"Spatial optimization of the sustainable aviation fuel supply chains from forest residues via fast pyrolysis/hydrotreatment considering feedstock ash content variability." **P. Li**, T. E. Yu, N. Labbé, N. Abdoulmoumine, M. Garcia-Perez, K. P. Hoyt, and B. C. English. *Biomass and Bioenergy*, 208 (2026), 108793.

"Assessing the impact of preprocessing and conversion technologies on the sustainable aviation fuel supply from forest residues in the Southeast USA." **P. Li**, T. E. Yu, C. Trejo-Pech, J. A. Larson, B. C. English, and D. N. Lanning. *Transportation Research Record*, 2678(9) (2024), 639–654.

"Beef price spread relationship with processing capacity utilization." C. Martinez, **P. Li**, C. N. Boyer, T. E. Yu, and J. G. Maples. *Journal of the Agricultural and Applied Economics Association*, 2(1) (2023), 133–145.

"Do green technology innovations contribute to carbon dioxide emission reduction? Empirical evidence from patent data." K. Du, **P. Li**, and Z. Yan. *Technological Forecasting and Social Change*, 146 (2019), 297–303.

TEACHING

Instructor: ECON 351 Monetary Economics (Spring 2025, 2026); ECON 322 The Global Economy: Trade and Development (Summer 2025); ECON 313 Intermediate Macroeconomics (Summer 2026).
Teaching assistant: Ph.D. Microeconomic Theory and Mathematical Methods (ECON 511, 512, 581); Agribusiness Operations Research (AREC 525).

HONORS

Charles B. Garrison Award for Excellence in Teaching, University of Tennessee	2026
Graduate Summer Fellowship, University of Tennessee	2025
J. Fred and Wilma Holly Award (best second-year paper), University of Tennessee	2024
J. Fred and Wilma Holly Fellowship, University of Tennessee	2023 – present